EXP NO:11 EXPERIMENT TO UNDERSTAND PANDAS LIBRARY USE CASES IN DATA SCIENCE.

CODE :

```
import numpy as np
import pandas as pd
list_data = [[1, 'Smith', 50000], [2, 'Jones', 60000]]
df = pd.DataFrame(list_data, columns=['Empd', 'Name', 'Salary'])
print(df)
print(df.info())
df = pd.read_csv("startups_600.csv")
print(df.info())
print(df.head())
print(df.tail())
df = pd.read_csv("employee_1000.csv")
print(df.head())
print(df.tail())
print(df.info())
print(df.salary)
print(type(df.salary))
print(df.salary.mean())
print(df.salary.median())
print(df.salary.mode())
print(df.salary.var())
```

```
print(df.salary.std())
print(df.describe())
print(df.describe(include='all'))
empCol = df.columns
print(empCol)
emparray = df.values
print(emparray)
employee_DF = pd.DataFrame(emparray, columns=empCol)
print(employee_DF)
```

OUTPUT :

```
  Empd   Name  Salary
0    1  Smith   50000
1    2  Jones   60000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
```

```
memory usage: 180.0+ bytes
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 5 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   R&D Spend       600 non-null    float64
 1   Administration  600 non-null    float64
 2   Marketing Spend  600 non-null    float64
 3   State           600 non-null    object
 4   Profit          600 non-null    float64
dtypes: float64(4), object(1)
memory usage: 23.6+ KB
None
   R&D Spend  Administration  Marketing Spend       State    Profit
0   33787.01       163739.48       478250.70  California  63753.42
1   55718.07        53688.04       368754.18     Florida  77770.43
2   35402.10        53318.53       176625.70     Florida  32413.62
3   17740.51        98541.53       148267.79  California  21578.94
4   24127.17       123296.48       174851.61    New York  49197.95
     R&D Spend  Administration  Marketing Spend       State     Profit
595  174307.36        62718.80       172938.73  California  171958.72
```

```
596  194697.79       157448.51        77229.87  California  149990.84
597  193775.57        60812.65        12723.88    New York  138755.39
598  149930.37        60688.51       322911.19    New York  147532.63
599   26017.25        51816.27       318451.16    New York   58240.98
  emp id                name  salary
0      1  SREYASKARI MULLAPUDI   48000
1      2     SREE VARSSINI K S   48000
2      3           ARUN KUMAR   90000
3      4              PRIYA S   90000
4      5              PRIYA S   75000
    emp id              name  salary
995    996           SWETHA R   90000
996    997          VIGNESH M   28000
997    998           ADITYA R   25000
998    999          ARUN KUMAR   30000
999   1000  SREE VARSSINI K S   28000
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  1000 non-null   int64
 1   name    1000 non-null   object
```

```
 2   salary  1000 non-null   int64
dtypes: int64(2), object(1)
memory usage: 23.6+ KB
None
0      48000
1      48000
2      90000
3      90000
4      75000
       ...
995    90000
996    28000
997    25000
998    30000
999    28000
Name: salary, Length: 1000, dtype: int64
<class 'pandas.core.series.Series'>
54982.0
50000.0
0    55000
Name: salary, dtype: int64
474538214.2142142
21783.898049114494
```

```
          emp id         salary
count  1000.000000    1000.000000
mean    500.500000   54982.000000
std     288.819436   21783.898049
min       1.000000   25000.000000
25%     250.750000   35000.000000
50%     500.500000   50000.000000
75%     750.250000   70000.000000
max    1000.000000  100000.000000
          emp id     name         salary
count   1000.000000     1000    1000.000000
unique         NaN       20           NaN
top            NaN  PRIYA S           NaN
freq           NaN       67           NaN
mean     500.500000      NaN   54982.000000
std      288.819436      NaN   21783.898049
min        1.000000      NaN   25000.000000
25%      250.750000      NaN   35000.000000
50%      500.500000      NaN   50000.000000
75%      750.250000      NaN   70000.000000
max     1000.000000      NaN  100000.000000
Index(['emp id', 'name', 'salary'], dtype='object')
[[1 'SREYASKARI MULLAPUDI' 48000]
```

```
 [2 'SREE VARSSINI K S' 48000]
 [3 'ARUN KUMAR' 90000]
 ...
 [998 'ADITYA R' 25000]
 [999 'ARUN KUMAR' 30000]
 [1000 'SREE VARSSINI K S' 28000]]
    emp id                name salary
0        1  SREYASKARI MULLAPUDI  48000
1        2     SREE VARSSINI K S  48000
2        3           ARUN KUMAR  90000
3        4              PRIYA S  90000
4        5              PRIYA S  75000
..     ...                  ...    ...
995    996            SWETHA R  90000
996    997           VIGNESH M  28000
997    998            ADITYA R  25000
998    999           ARUN KUMAR  30000
999   1000     SREE VARSSINI K S  28000

[1000 rows x 3 columns]
```